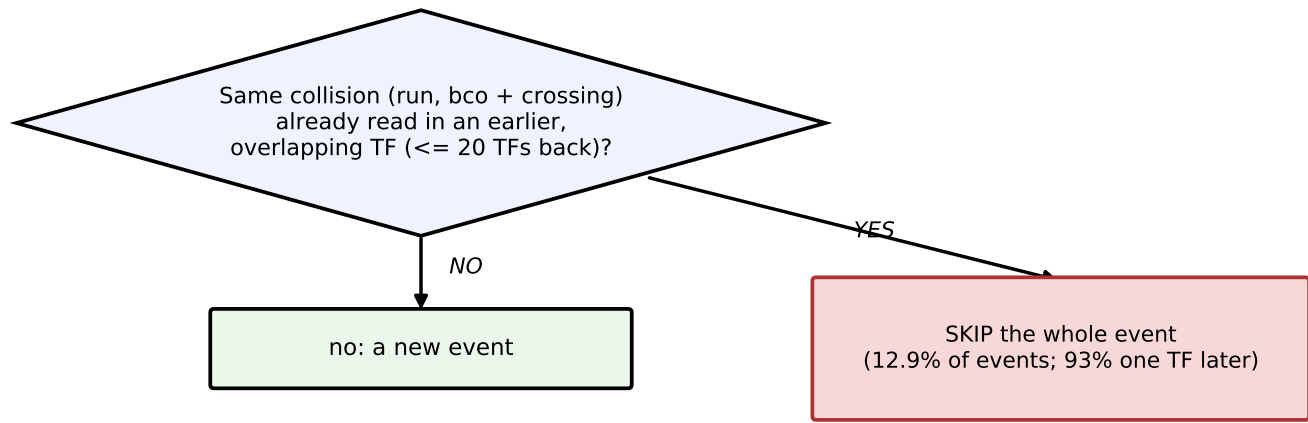


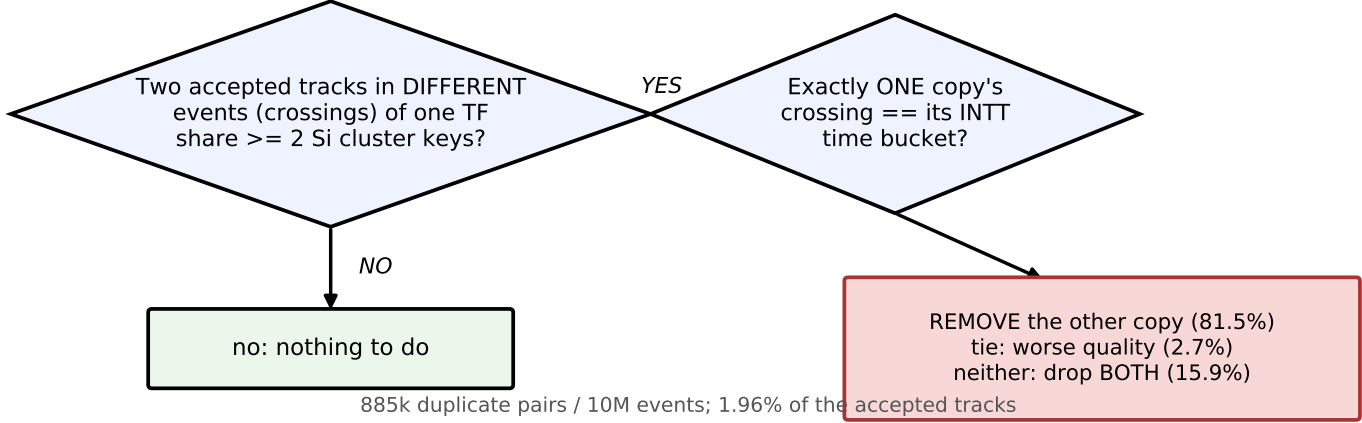
Split-track and duplicate treatment, ana573 default, page 1: events and tracks -- every decision in processing order

STAGE 0 -- pre-passes over all events, before the event loop (event = one crossing with a vertex; the DSTs hold many events per trigger frame, TF)

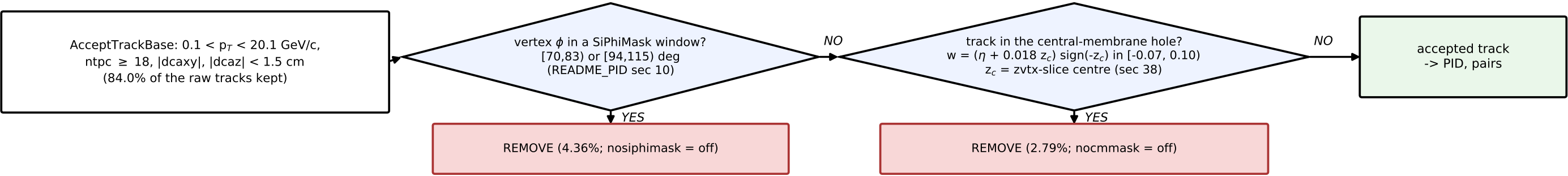
(a) BuildTFDupRows (noTFdup = off)



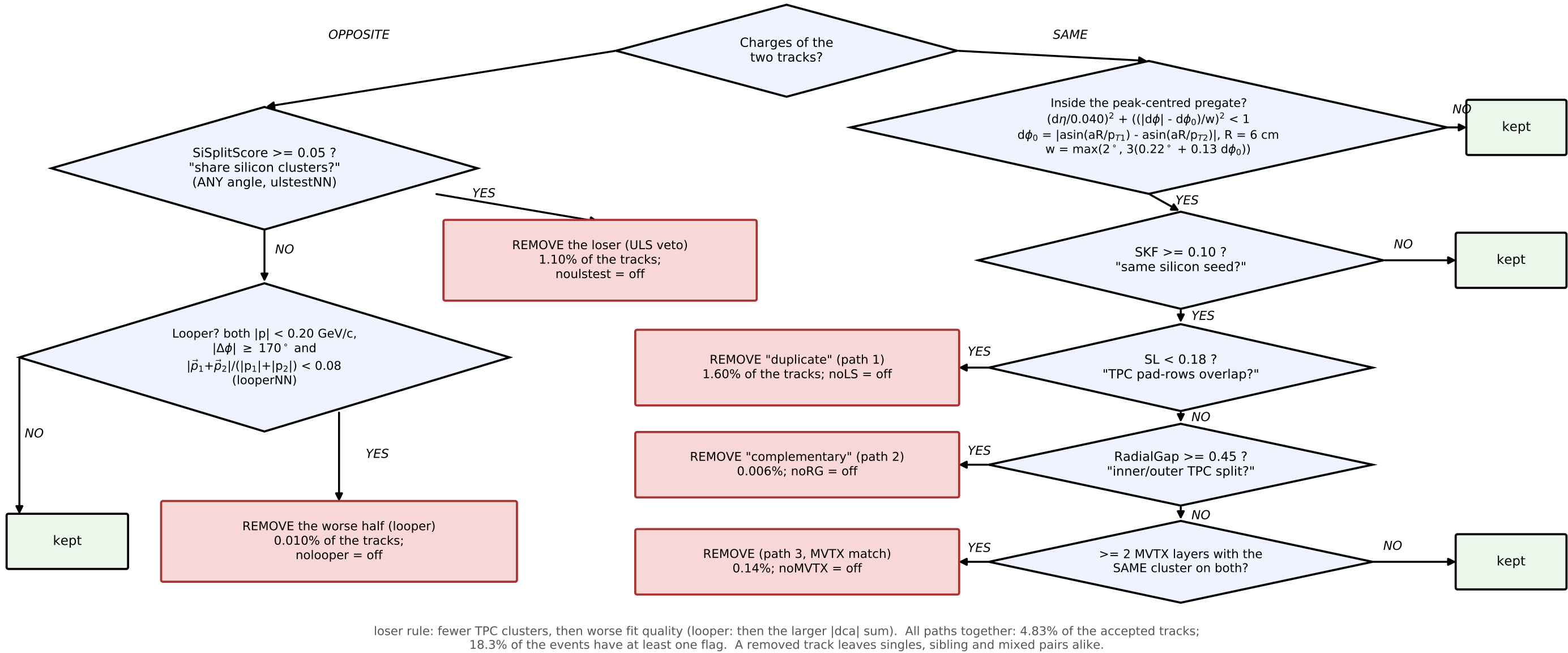
(b) BuildXTFLosers (noXTF = off, XTFclean = non-strict)



STAGE 1 -- per track (AcceptTrack)

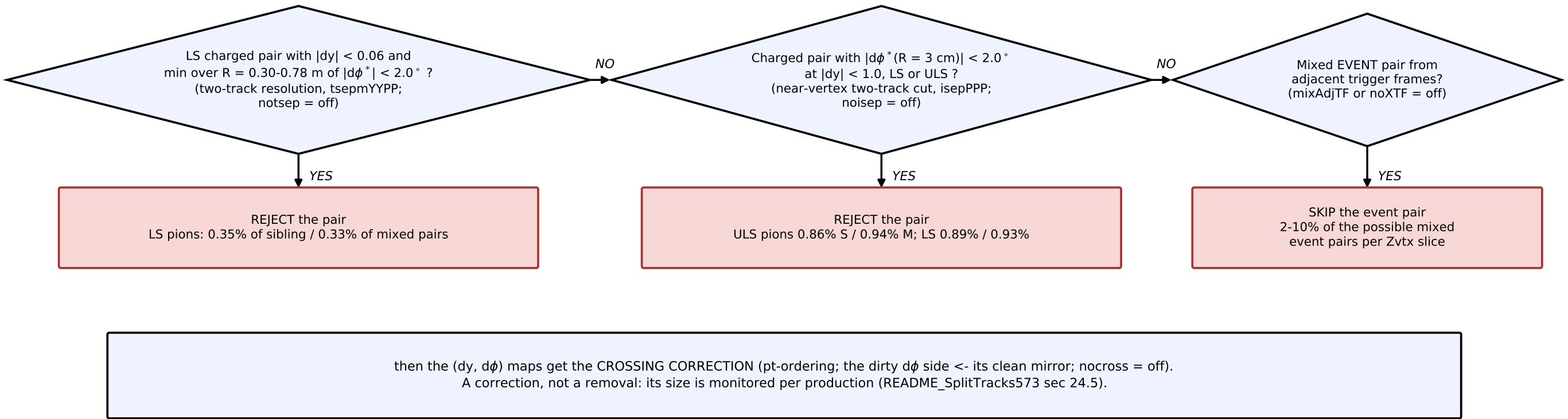


STAGE 2 -- per event (one crossing), every pair of accepted charged tracks (corral_loop.cxx split-track pre-pass)



Opposite-sign candidates are every accepted charged track (not only pions); daughters of V0s in their mass peak stay out; Lambda/Lbar daughters (not K0S) are split partners (page 3). Pair-level cuts, mixing, the crossing correction and the watch items: page 2. Pictures of each case: ana573/SplitTrackTypes_ana573.pdf.

STAGE 3 -- pair level (CalcRm::PairInfo): sibling AND mixed pairs alike, in this order; then the maps



WATCH -- monitored in every production, not removed (user's rule: monitor, do not fix)

The $\pm 15^\circ$ ULS notch (secs 32-34)
a $\sim 4\%$ deficit of ULS SIBLING pion pairs at $10\text{-}20^\circ$ vertex $d\phi$ ($\sim 12\%$ in R2 at $|dy| < 0.1$). Cause: pairs whose tracks MEET in ϕ at $R_m \approx 0.10\text{-}0.30$ m (INTT-TPC gap) are lost in the silicon-TPC matching; no Corral cleaner makes it. For tracking.
Page: "The $\pm 15^\circ$ ULS notch: WATCH" (S/M vs $d\phi$, R_m , p_T , N_{ch}).

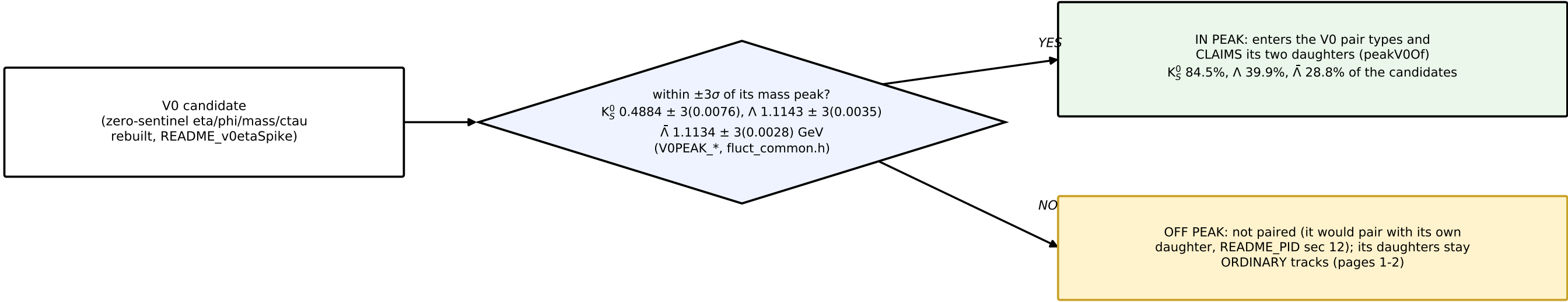
Also monitored

- the crossing losses of each production (ana573 50M, sec 24.5: $+5.6\text{e-}3$ pairs/event $\pi^+\pi^-$, $3.7\text{-}10\times$ ana532)
- the same-TF share of mixed pairs (5.8%, sec 18.20)
- V0 masses and C(Q) peak positions (momentum scale: a few % in the field fixes them)
- tracks removed per cleaner: the split-removal page and the log

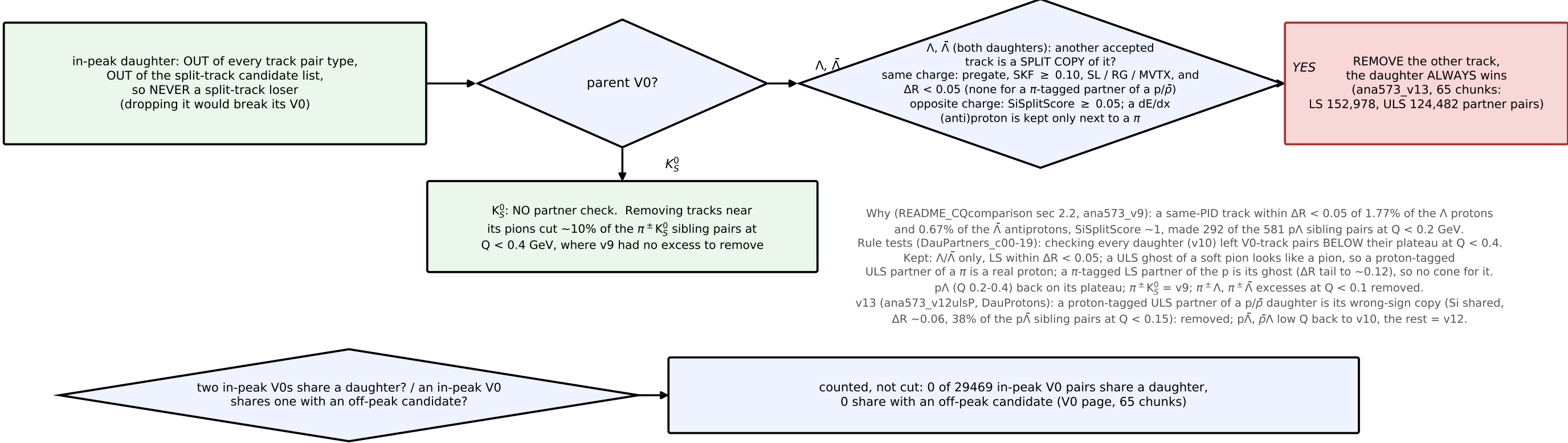
For tracking, beyond the notch: the π^+/π^- split asymmetry (secs 15.2, 19); the SiPhiMask zones, the 90° pattern of $\langle |dcaxy| \rangle$ vs ϕ and the dcaz ramp in η (README_PID sec 10).

Split-track and duplicate treatment, ana573 default, page 3: V0s -- which V0s pair, and what happens to their daughters

STAGE V1 -- per event, every KFP V0 candidate (K_S^0 , Λ , $\bar{\Lambda}$; candidate cuts upstream: README_settingKFPcuts)

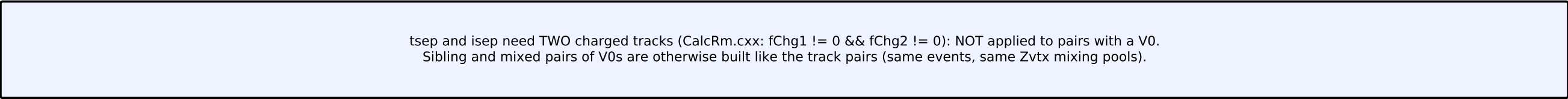


STAGE V2 -- per event, the in-peak daughters in the track machinery (corral_loop.cxx)



Why (README_CQcomparison sec 2.2, ana573_v9): a same-PID track within $\Delta R < 0.05$ of 1.77% of the Λ protons and 0.67% of the $\bar{\Lambda}$ antiprotons, SiSplitScore ~ 1 , made 292 of the 581 p Λ sibling pairs at $Q < 0.2$ GeV. Rule tests (DauPartners_c00-19): checking every daughter (v10) left V0-track pairs BELOW their plateau at $Q < 0.4$. Kept: $\Lambda/\bar{\Lambda}$ only, LS within $\Delta R < 0.05$; a ULS ghost of a soft pion looks like a pion, so a proton-tagged ULS partner of a π is a real proton; a π -tagged LS partner of the p is its ghost (ΔR tail to ~ 0.12), so no cone for it. p Λ (Q 0.2-0.4) back on its plateau; $\pi^\pm K_S^0 = v9$; $\pi^\pm \Lambda$, $\pi^\pm \bar{\Lambda}$ excesses at $Q < 0.1$ removed. v13 (ana573_v12ulsP, DauProtons): a proton-tagged ULS partner of a p/ $\bar{p}$ daughter is its wrong-sign copy (Si shared, $\Delta R \sim 0.06$, 38% of the p $\bar{\Lambda}$ sibling pairs at $Q < 0.15$): removed; p $\bar{\Lambda}$, $\bar{p}\Lambda$ low Q back to v10, the rest = v12.

STAGE V3 -- pair level (CalcRm::PairInfo), V0-V0 and track-V0 pairs, sibling and mixed alike



WATCH -- monitored, not removed

